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SEAT No. _____

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SARDAR PATEL UNIVERSITY

Master of Science (information Technology - Semester II)(NC) Examination - 2022
PS02CINT52 – SOFTWARE ENGINEERING

DATE: 10/11/2022, Thursday

TIME: 10:00 A.M. TO 01:00 P.M.



TOTAL MARKS: [70]

Q.1 Choose the most appropriate option for the below multiple choice questions. [08]

1. _____ model is a simplest and most widely used model, in which phases are arranged in linear order.
(A). Waterfall (B). Iterative (C). Spiral (D). Prototype
2. _____ model eliminates the 3rd limitation of waterfall model.
(A). Spiral (B). Iterative (C). Prototype (D). None of these
3. _____ provides as a reference for validation of the final product.
(A). Analysis Report (B). Test Report (C). SRS (D). None of these
4. _____ requirement focus on the functionality of the software components that build the system.
(A). Design (B). Domain (C). Non-functional (D). Functional
5. _____ is also known as divide and conquer technique
(A). Problem Partitioning (B). Top-Down Approach
(C). Abstraction (D). None of these
6. UML stands for _____.
(A). United Modeling Language (B). Unified Modeling Language
(C). Universal Modeling language (D). Unique Modeling Language
7. A human action that results in a software failure is known as _____.
(A). Defect (B). Failure (C). Error (D). None of these
8. _____ is known as functional testing.
(A). Enhanced (B). State based (C). Whitebox (D). Blackbox

Q.2 Answer the following questions in short. (Attempt any seven) [14]

1. Define Software Engineering. List the goals of Software Engineering.
2. Why we need SRS?
3. Write a note on WBS [Work Breakdown Structure].
4. List out the components of SRS.
5. Write a short note on Object Oriented Analysis and Design.
6. Write a short note on Sequence Diagram.
7. Differentiate: Blackbox Testing & Whitebox Testing.
8. Define Error, Fault & Failure.

①

C.P.T.O.

9. List 3 common coding errors.

Q.3 Answer the following questions in detail.

[A] Write a note on Waterfall Model in detail. [06]

(B) Write a note on Iterative Model in detail. [06]

OR

(B) List the characteristics of Software Process. Explain it in detail. [06]

Q.4 Answer the following questions in detail.

(A) Write a note on COCOMO model in detail. [06]

(B) What is SRS? List the characteristics of SRS. Explain it in detail. [06]

OR

(B) Write a note on Risk Management. [06]

Q.5 Answer the following questions in detail.

(A) What is Coupling? Explain types of coupling in detail. [06]

(B) What is Design in Software Engineering? Explain Design Principles in detail. [06]

OR

(B) What is Cohesion? Write a note on different levels of cohesion. [06]

Q.6 Answer the following questions in detail.

(A) What is Software Testing? Explain Blackbox [Functional] Testing in detail. [06]

(B) Write a note on TDD [Test Driven Development] [06]

OR

(B) Write a note on Alpha & Beta Testing. [06]

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(2)